

Expected Outcome: Projects are expected to contribute to the following outcomes:

- Increase the awareness of Higher Education Institutions and research organisations to conventional and non-conventional threats
- Boost the capacities of Higher Education Institutions and research organisations to deter and respond to non-conventional threats
- Improve the resilience of Higher Education Institutions and research organisations

Scope: The EU Higher Education Institutions and research organisations continue to face serious threats, such as to academic freedom, freedom of research and institutional autonomy, as well as increasing distrust in science and scientific experts, next to various forms of foreign interference. In addition, acute threats increasingly take non-conventional forms, such as cyber-attacks, infiltration or disinformation campaigns. These actions seek to access research findings, destabilise and endanger our higher education and research institutions and undermine our fundamental European values. To respond to these threats, the EU Higher Education Institutions and research organisations need to boost significantly their capacities in detecting and tackling these threats and they need to be well informed and geared with simple and effective tools ensuring security while respecting the openness of research and innovation. Given the very nature of the topic, continued exchange with member states and relevant stakeholders is necessary in order to expand the knowledge of conventional and non-conventional threats to Higher Education Institutions and research organisation, and develop appropriate actions.

Proposals are expected to: (i) analyse possible threats and identify mitigation measures; (ii) increase awareness of threats within higher education institutions and research organisations; (iii) promote networking and optimising of infrastructure and resources to ensure adequate preparedness and prevention at European level.

In the framework of the new ERA, this action should also complement and reinforce EU action for academic freedom as well as the European *Pact for Research and Innovation*, which aims at deepening the implementation of the new ERA's objectives, setting out commonly agreed values and principles and indicating the areas where Member States will jointly develop priority actions.

CITIZEN SCIENCE

Proposals are invited against the following topic(s):

HORIZON-WIDERA-2021-ERA-01-60: A capacity-building and brokering network to make citizen science an integral part of the European Research Area

Specific conditions	
<i>Expected EU contribution per</i>	The Commission estimates that an EU contribution of around EUR 4.00 million would allow these outcomes to be addressed appropriately. Nonetheless, this does not preclude submission and selection of a

Horizon Europe - Work Programme 2021-2022
Widening participation and strengthening the European Research Area

<i>project</i>	proposal requesting different amounts.
<i>Indicative budget</i>	The total indicative budget for the topic is EUR 4.00 million.
<i>Type of Action</i>	Coordination and Support Actions
<i>Eligibility conditions</i>	The conditions are described in General Annex B. The following exceptions apply: Legal entities established in non-associated third countries may exceptionally participate in this Coordination and support action.
<i>Legal and financial set-up of the Grant Agreements</i>	The rules are described in General Annex G. The following exceptions apply: Beneficiaries will be subject to the following additional dissemination obligations: Beneficiaries must make proactive efforts to freely share, in a timely manner and as appropriate, all relevant results with the other grants awarded from this work programme part subject to the same additional dissemination obligations. Beneficiaries must acknowledge and incorporate these obligations in the proposal, outlining the efforts they will make towards meeting them, and in Annex I to the Grant Agreement.

Expected Outcome: Projects are expected to contribute to the following expected outcomes:

- Strengthened links and collaboration between citizen science initiatives and other research and innovation actors;
- Increased capacity to conduct excellent research and innovation through citizen science, while maximising other potential benefits of citizen science;
- Data infrastructures better aligned to the needs of citizen science, and improved data practices employed by citizen science initiatives;
- Europe positioned as a leader in citizen science throughout the entire research and innovation system with flourishing and mutually beneficial global collaborations.

These targeted outcomes in turn contribute to medium and long-term impacts:

- Increased engagement of citizens in research and innovation;
- Increased public trust in research and innovation;
- Increased capacity in the EU R&I system to conduct open science and to set it as a *modus operandi* of modern science;
- Increased openness of the EU R&I system.

Scope: Citizen Science is a rapidly emerging mode of research and innovation that shows huge promise in terms of collecting new qualities and quantities of data, harnessing collective intelligence, improving science-society literacy, and improving the relationship between science and society. However, it is embedded institutionally only to a limited extent, at an early stage of developing data infrastructures, and lacks the capacities and sustainable resourcing required to live up to its potentials.

This action will become a central point of exchange between citizen science in Horizon 2020, Horizon Europe, and other EU-level and sub-national programmes and initiatives. It should amplify significant outcomes of citizen science in areas such as Horizon Europe's Missions, should they be confirmed, Clusters and Partnerships, the Sustainable Development Goals, and the Green Deal, across all parts of the research and innovation system. It should maintain up-to-date repositories of initiatives, good practices and tools, and become a stage for discussion and collaboration.

It should offer extensive *in situ* brokering, skills, training, and capacity building services to citizen science practitioners, civil society, public authorities, businesses/SMEs, formal and informal education establishments, and research funding and performing organisations, with a view to raising their awareness, knowledge and skills to collaborate with, support, and implement citizen science.

Co-ordination of and support for citizen science to work towards FAIR (and in many cases open) data should cut across activities. The action should become a key interlocutor between citizen science initiatives and existing thematic databases and infrastructures. It should support - and 'connect the dots' between - existing efforts to make citizen science data FAIR and open, make new efforts to liaise between citizen science and infrastructures where they are needed, identify unmet needs, and develop policy recommendations. Overall the action should move citizen science towards open science as its *modus operandi*.

Underlying these activities should be efforts to reduce disparities in awareness and actual practice of citizen science across disciplines in the ERA (e.g. at local, regional, national and EU levels). Moreover, significant efforts should be made to be inclusive in citizens' involvement in terms of geography, gender, ethnicity, disability, age, socio-economic background etc. In order to achieve the expected outcomes, international cooperation and mutual learning that promise tangible scientific, societal or policy impacts is advised.

The action should build on and valorise the results of earlier projects in the Science and Society (FP6), Science in Society (FP7) and Science with and for Society (Horizon 2020) programmes, in particular projects focused on public engagement, responsible research and innovation, and citizen science, as well as of national and regional initiatives, and should aim to provide a seamless transition between previous supporting actions and this new action.

The action should be no shorter than 3 years.